GHG Protocol Corporate Accounting and Reporting Standard

World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD)

Chapter 4: Setting Operational Boundaries - Scopes 1, 2, and 3

The GHG Protocol defines three operational accounting scopes to categorize direct and indirect emission sources:
Scope 1 Direct GHG Emissions: Emissions from operations owned or controlled by the organization, including combustion of fossil fuels (diesel, natural gas, petrol) in stationary equipment such as generators and furnaces.
Scope 2 Electricity Indirect Emissions: Emissions resulting from the generation of purchased electricity consumed by the company. Scope 2 accounting requires reporting location-based grid average emission factors published by regional electrical authorities.
Scope 3 Other Indirect Emissions: An optional reporting category that accounts for all other indirect emissions resulting from organization activities from sources not owned or controlled, notably upstream freight and distribution.

Chapter 6: Calculating GHG Emissions and Activity Data

Emissions must be calculated through the fundamental equation:
GHG Emissions (kg CO_2e) = Activity Data x Emission Factor
Activity data measures the magnitude of human activity resulting in greenhouse gas emissions (e.g., litres of diesel fuel consumed, kilowatt-hours of grid electricity used, or tonne-kilometres of freight transported). Emission factors quantify the mass of greenhouse gas emitted per unit of activity.

Chapter 9: Managing Inventory Quality and Uncertainty Ranges

Emissions estimations inherently carry scientific and measurement uncertainty. In manufacturing environments with imperfect records, uncertainty ranges (low and high bounds) must be explicitly quantified to account for equipment maintenance degradation, load variations, and fuel quality tolerances. Reports must clearly declare that emissions figures are estimates subject to stated assumptions and do not constitute formal third-party regulatory certification.